SSW 590 — DevOps

Group 1 DevOps IT Manual

Version 0.1

Draft: Gitea Chapter in Progress

by

Group 1: Brahmi Dalvi and Sakshi Deshpande

Stevens Institute of Technology

October 12, 2026

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Template Attribution

This manual is built from the SSW 590 DevOps assignment template (version 1.3, distributed through Canvas), developed by **Darian Muresan, Ph.D.**, in conjunction with **OpenAI Codex** and **Anthropic Claude Code**.

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Group 1: Brahmi Dalvi and Sakshi Deshpande
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This is Group 1's living DevOps IT manual for SSW 590. It records what the group actually built, tested, and learned, one chapter per course module. The first chapter documents the startup's Gitea work hub: a student-pack domain, a version-pinned Gitea deployment, HTTPS with automated certificate renewal, a minimal firewall exposure, and an issue-to-reviewed-merge team workflow. Items not yet executed are marked as pending; no result is claimed before it is run.

Document History

How to Use This Chapter

Add the newest approved change at the top. Retain every approved entry and link each entry to the changed chapter, table, figure, or controlled artifact. Every entry must say whether the Hosts and Passwords tables changed. Author initials: BD = Brahmi Dalvi, SD = Sakshi Deshpande, CC = Claude Code (AI assistant; see the AI Prompt Log).

Table 1: Document Update History

Date / Version	Author Initials	Change	Changed At
October 12, 2026 / 0.1	BD, CC	Initial setup: created the separate IT manual from the course Canvas template (version 1.3, one chapter per file, template guides retained as appendices), the Gitea chapter outline with the eight rubric-aligned subsections and an extra-credit subsection, and the Hosts and Passwords tables. Hosts and Passwords tables created, with no real hosts or secrets recorded yet. No infrastructure exists yet; every evidence item is pending.	Chapter 1, Chapter 2, Chapter 3

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Chapter 1

Introduction

1.1 Purpose

This is Group 1's operational IT manual for the SSW 590 startup. It is a living record: one chapter is added per delivered capability, and the compiled PDF is the only artifact submitted for each assignment. It is separate from the earlier assignment report in the `report/` folder of the repository.

1.2 Team and Repository

Item	Value
Group	Group 1
Members	Brahmi Dalvi (BD), Sakshi Deshpande (SD)
Course repository	https://github.com/brahmi9811/ssw-590-group1
Team Gitea	**[TODO: link once git.example.com is live]**

1.3 Chapter Map

Where	Content
Chapter 2	Hosts and Passwords tables (hints only)
Chapter 3	Startup foundation with Gitea (course Chapter 2)

1.4 Repository Layout

```
assignment-4/manual/   this LaTeX manual (source stays in the team; submit only the PDF)
assignment-4/gitea/    versioned, non-secret deployment files (compose, proxy config)
```

```
assignment-4/*.md      work split and per-person instructions
report/                earlier submitted assignment (do not modify)
```

Chapter 2

Hosts and Passwords

Keep both tables current; a missing or stale table costs 10 points. The Passwords table holds *hints and vault references only*: never a password, derivation key, recovery code, token, or private key.

2.1 Hosts

Table 2.1: Hosts

Hostname	Role	Provider	Address / DNS	Exposed ports	Owner / HTTPS URL / backup
[TODO: git host]	Gitea + TLS proxy	[TODO: AWS or DO]	[TODO: IP, DNS]	[TODO: 443, 80?, admin SSH]	[TODO: owner; https://git.example.com; none yet]

2.2 Passwords (hints only)

Table 2.2: Passwords (hints only)

Service / account	Owner	Vault reference or hint (no secret)	Privilege / MFA
[TODO: Gitea admin]	[TODO:]	[TODO: where the real credential lives]	[TODO: admin / MFA?]

Chapter 3

Startup Foundation with Gitea

3.1 Overview and Status

This chapter answers the course manual's Chapter 2 [1]: the startup's Gitea work hub with a student domain, HTTPS, minimal public exposure, and persistent data. The eight numbered subsections below match the eight rubric rows (100 points) in the same order. Optional backup and restore evidence is a separate extra-credit subsection and is not part of the eight; the course treats it as optional, but Group 1 has committed to completing it.

Provider status

The assignment specifies a DigitalOcean Droplet. Group 1 has asked the instructor to allow AWS EC2 instead; **no reply has been received**, so no cloud resource has been created. Replace this note with the instructor's answer once received. **[TODO: record the instructor's decision and date]**

Table 3.1: Rubric rows, owners, and status

Subsection	Points	Owner	Status
Student Pack domain and DNS	10	BD	PENDING
Pinned Gitea deployment	10	BD	PENDING
Database and persistence	10	SD	PENDING
HTTPS and certificate renewal	15	BD	PENDING
Firewall and external exposure	15	BD	PENDING
Startup team workflow	20	SD	PENDING
Architecture and operation	5	SD	PENDING
Manual quality and controls	15	BD	PENDING

Subsection	Points	Owner	Status
Extra credit: backup and restore (committed)	up to 20	SD	PENDING

3.2 Required Subsections

3.2.1 Student Pack domain and DNS (10 points)

Owner: BD. PENDING No evidence recorded yet.

Required content (from the course manual):

- Domain offer, eligibility, first-year cost, renewal cost, expiration
- Chosen Gitea hostname and clickable HTTPS link
- DNS record screenshot and an independent lookup result (e.g. `nslookup` against 8.8.8.8, run by a teammate)
- How a future team tracks renewal (calendar owner, date)

[TODO: write the explanation, then add dated, labeled, sanitized evidence (expected vs. actual). A screenshot alone does not replace an explanation. Never claim a test that was not run.]

3.2.2 Pinned Gitea deployment (10 points)

Owner: BD. PENDING No evidence recorded yet.

Required content (from the course manual):

- Droplet/EC2 and Gitea image versions, image source (Docker Hub `gitea/gitea`)
- Deployment commands, restart policy, persistent mounts, proxy connection
- Redacted Compose excerpt and link to `gitea/docker-compose.yml` in the team repository
- Steps another team could repeat

[TODO: write the explanation, then add dated, labeled, sanitized evidence (expected vs. actual). A screenshot alone does not replace an explanation. Never claim a test that was not run.]

3.2.3 Database and persistence (10 points)

Owner: SD. PENDING No evidence recorded yet.

Required content (from the course manual):

- **Decision (final, team, October 12, 2026): SQLite.** Record the settings, the database-file path under the persistent `/data` volume, the volume mount, and why no database secret exists. Justify: two-person team, no extra container or credential, simple consistent backup; note it may not suit higher concurrency
- Repository and issue identifiers created before the test

- Exact container-replacement command (volumes kept), before/after evidence, conclusion
- Not applicable: PostgreSQL is not used, so there is no database port, healthcheck, or database secret to document

[TODO: write the explanation, then add dated, labeled, sanitized evidence (expected vs. actual). A screenshot alone does not replace an explanation. Never claim a test that was not run.]

3.2.4 HTTPS and certificate renewal (15 points)

Owner: BD. PENDING No evidence recorded yet.

Required content (from the course manual):

- Comparison of two certificate options and the justification for the chosen one
- Where TLS terminates; how private traffic reaches Gitea
- Valid hostname certificate: issuer and expiry
- HTTP-to-HTTPS redirect behaviour
- Renewal automation plus a successful renewal test or dry run
- Redacted proxy configuration and test output

[TODO: write the explanation, then add dated, labeled, sanitized evidence (expected vs. actual). A screenshot alone does not replace an explanation. Never claim a test that was not run.]

3.2.5 Firewall and external exposure (15 points)

Owner: BD. PENDING No evidence recorded yet.

Required content (from the course manual):

- Cloud firewall / security group name and attachment, inbound rules, source ranges, IPv4 and IPv6
- Host and Compose port mappings
- Small network diagram
- External test (source, time, expected vs. actual) for 443, 80, admin SSH, Gitea backend 3000, database port
- Corrections made; why HTTPS does not replace a firewall

[TODO: write the explanation, then add dated, labeled, sanitized evidence (expected vs. actual). A screenshot alone does not replace an explanation. Never claim a test that was not run.]

3.2.6 Startup team workflow (20 points)

Owner: SD. PENDING No evidence recorded yet.

Required content (from the course manual):

- Links to the organization and product repository; README
- Two members and their roles
- At least five actionable issues (product and operations)

- Two issues traced through linked branch/PR, peer review, merge, closure (needs both members)

[TODO: write the explanation, then add dated, labeled, sanitized evidence (expected vs. actual). A screenshot alone does not replace an explanation. Never claim a test that was not run.]

3.2.7 Architecture and operation (5 points)

Owner: SD. **PENDING** No evidence recorded yet.

Required content (from the course manual):

- Architecture diagram: end-to-end request and data paths
- Where data lives; which interfaces are public vs. private
- How the team restarts and troubleshoots the service
- Persistence test is not a backup/restore test

[TODO: write the explanation, then add dated, labeled, sanitized evidence (expected vs. actual). A screenshot alone does not replace an explanation. Never claim a test that was not run.]

3.2.8 Manual quality and controls (15 points)

Owner: BD. **PENDING** No evidence recorded yet.

Required content (from the course manual):

- Increase the manual version; dated history entries for initial setup and this update, linking sections and evidence
- Update Hosts and Passwords tables (hints only) and mention both in the history
- Headings, captions, citations, glossary/index entries; secrets redacted
- State any failed tests and corrective actions

[TODO: write the explanation, then add dated, labeled, sanitized evidence (expected vs. actual). A screenshot alone does not replace an explanation. Never claim a test that was not run.]

3.3 Extra Credit

3.3.1 Extra credit: verified backup and restore (up to 20 extra-credit points; committed by the team)

Owner: SD. **PENDING** No evidence recorded yet.

Required content (from the course manual):

- Backup method, coverage, storage location, time (5)
- Sanitized backup log/listing; DB and app data form one recoverable state (5)
- Test repository and issue present and usable in a separate temporary restored instance, with restore log and screenshots (10)
- Do not overwrite the working service; do not include archives, dumps, or keys

[TODO: write the explanation, then add dated, labeled, sanitized evidence (expected vs. actual). A screenshot alone does not replace an explanation. Never claim a test that was not run.]

Bibliography

[1] D. Muresan, *DevOpsStartup: A Startup DevSecOps Assignment Manual, Version 1.18, Student Edition*, Stevens Institute of Technology, 2026, sSW 590 course manual (PDF, dosManual-V1.18.pdf); Chapters 1 and 2 govern this chapter.

[2] Gitea, "Installation with docker," https://docs.gitea.com/installation/install-with-docker/, cited from the course manual bibliography; re-verify before relying on it.

[3] ——, "Configuration cheat sheet," https://docs.gitea.com/administration/config-cheat-sheet/, cited from the course manual bibliography; re-verify before relying on it.

[4] Caddy, "Automatic https," https://caddyserver.com/docs/automatic-https, cited from the course manual bibliography; re-verify before relying on it.

[5] DigitalOcean, "How to configure firewall rules," https://docs.digitalocean.com/products/networking/firewalls/how-to/configure-rules/, cited from the course manual bibliography; re-verify before relying on it.

[6] Docker, "Packet filtering and firewalls," https://docs.docker.com/engine/network/packet-filtering-firewalls/, cited from the course manual bibliography; re-verify before relying on it.

[7] Let's Encrypt, "Challenge types," https://letsencrypt.org/docs/challenge-types/, cited from the course manual bibliography; re-verify before relying on it.

[8] GitHub Education, "Github student developer pack: Domain offers," https://education.github.com/pack?sort=az&tag=Domains, offer eligibility and terms must be checked at the time of use.

Appendix A

Document Authoring Guide

A.1 How to Use This Appendix

Use this appendix to maintain the template's file conventions, build process, and review rules. Update it when the document structure or toolchain changes.

A.2 Template Provenance

This template was developed by **Darian Muresan, Ph.D.**, in conjunction with **OpenAI Codex** and **Anthropic Claude Code**. Keep this attribution distinct from the assignment-specific instructor metadata.

A.3 File Naming and Responsibilities

Every chapter file shall use the `devops` prefix, a PascalCase chapter name, and the `.tex` suffix. Each such file shall contain one complete chapter, beginning with `\chapter{...}`, and all of its sections and subsections. Do not split one chapter into section-level `.tex` files. For example, all of Chapter 2 is contained in `devopsExampleAssignment.tex`. Do not add requirements or architecture-design chapters; this template records a DevOps learning activity and its operational evidence.

A.4 Synonyms and Controlled Terms

Use one preferred term for each important concept. If an assignment uses equivalent terms, record them here before release. For example, state whether "continuous delivery" refers to continuous delivery or continuous deployment (CD) generally or specifically excludes automatic production deployment. Add important terms to `devopsGlossary.tex` and index them where used.

A.5 Building the PDF

The template requires a LaTeX distribution with `pdflatex`, `bibtex`, `makeglossaries`, and `makeindex`. Install the tools using the platform instructions below, then run one of the following platform builds.

A.5.1 macOS and Linux Build

```
make -f makefileMacLinux pdf
```

A.5.2 Windows PowerShell Build

```
.\makefilePowerShell.ps1 -Action pdf
```

The completed build regenerates references, bibliography, glossary, and index. Review the log and visually inspect affected PDF pages. Do not edit generated auxiliary files or the PDF directly.

A.6 Install LaTeX Tools on macOS

A.6.1 Recommended Installation

Install the current full MacTeX distribution. It includes TeX Live, `pdflatex`, `bibtex`, `latexmk`, `makeglossaries`, `makeindex`, and useful applications such as TeXShop and TeX Live Utility. The full distribution is the least troublesome choice for this template because it includes the packages used by the glossary, table, listing, color, and hyperlink features.

Either download the signed installer from the MacTeX web site and complete the macOS installation dialog, or, if Homebrew is already installed, run:

```
brew install --cask mactex
```

Close and reopen Terminal after installation. If the TeX commands are not found, add the MacTeX binary directory to the current shell session:

```
export PATH="/Library/TeX/texbin:$PATH"
```

Do not place this command in a shell profile until it has been verified on the local computer.

A.6.2 Verification and Compilation

From the template directory, verify the tools:

```
command -v pdflatex bibtex latexmk makeglossaries makeindex
```

Each command should print a path. Then create the PDF:

```
make -f makefileMacLinux pdf
```

To remove generated intermediate files while keeping the PDF, run:

```
make -f makefileMacLinux clean
```

A.7 Install LaTeX Tools on Windows

A.7.1 Recommended Installation

Install the current Basic MiKTeX distribution using the official MiKTeX Windows installation guide. During installation, choose a per-user installation when appropriate and set missing-package installation to *Ask me first* or *Yes*. The latter is convenient for a student machine with Internet access; the former makes each package installation explicit.

Open MiKTeX Console after installation, apply available updates, and install the `latexmk` package if the verification command below cannot find it. On an organization-managed computer, obtain approval before installing software or changing execution policy.

A.7.2 Verification and Compilation

Open a new PowerShell window in the template directory and verify:

```
Get-Command pdflatex,bibtex,latexmk,makeglossaries,makeindex
```

Each command should display a command record. Build the PDF with:

```
.\makefilePowerShell.ps1 -Action pdf
```

If PowerShell blocks the script under a managed execution policy, ask the system administrator for the approved procedure. Do not bypass institutional security policy. To remove intermediate build files, run:

```
.\makefilePowerShell.ps1 -Action clean
```

A.8 Common Build Problems

Symptom	Resolution
Command not found	Close and reopen the terminal. On macOS, verify the TeX binary path; on Windows, verify MiKTeX completed installation and is on the user PATH.
Missing LaTeX package	Use MiKTeX Console or its approved on-the-fly package installation prompt on Windows. On macOS, install the full MacTeX distribution or use TeX Live Utility to add the missing package.
References or glossary incomplete	Run the supplied platform build instead of a single `pdflatex` command; it runs BibTeX, the glossary tool, and the index tool in the required order.
Script is blocked	Follow the institution's approved PowerShell policy or run the documented build commands with an approved administrator.

A.9 Authoring and Review Rules

Replace bracketed placeholders with assignment facts. Keep operational examples safe, bounded, and free of credentials. Link activities to evidence, retain history, and state unknowns or instructor decisions explicitly. Rebuild after structural, cross-reference, table, glossary, or index changes.

Appendix B

AI Agent Usage Guide

B.1 How to Use This Appendix

An AI agent that creates, reviews, or maintains an assignment shall read this file first. It shall identify the task in the routing table, read the minimum authoritative sources, and expand context only for a discovered dependency or inconsistency. This guide is maintained in LaTeX so it is visible, versioned, and reviewable with the assignment.

B.2 Governing Principles

The authored LaTeX files, versioned automation, and approved assignment materials are authoritative. The PDF is a review artifact and shall not be edited directly. AI may draft, explain, analyze, check consistency, and assist with builds. It shall not invent completion, evidence, access, costs, grades, approvals, command output, security posture, or operational results.

B.3 Chapter File Convention

Every assignment chapter shall be stored in exactly one file named with the `devops` prefix, a PascalCase chapter name, and the `.tex` suffix. For example, Chapter 2 is fully contained in `devopsExampleAssignment.tex`. A chapter file shall begin with its `\chapter{...}` command and contain all of that chapter's sections and subsections. Do not split one chapter across files, create section-only `.tex` files, or use `\input` or `\include` to assemble a chapter from smaller fragments.

Supporting non-chapter files may use a different name only when they are not chapter content: `prologue.tex`, `abstract.tex`, `devopsGlossary.tex`, and `devopsReferences.bib` are examples. Appendix chapters follow the same chapter-file convention.

B.4 AI Quick Start

1. Read this guide and identify the requested outcome.
2. Select the closest task below and load only its listed sources.
3. Inspect supplied facts and report consequential gaps or assumptions.
4. Make narrowly scoped changes; preserve stable IDs, glossary terms, and history.
5. Never request, print, store, or commit credentials, tokens, personal data, or production data.
6. Build the complete PDF, inspect the log, visually inspect affected pages, and record material AI use in Chapter C.

B.5 Task Routing

Table B.1: AI Task-Routing Table

Task	**Minimum authoritative sources**
Initialize an assignment	`devopsAssignment.tex`, `prologue.tex`, `devopsIntroduction.tex`, `devopsExampleAssignment.tex`, and the instructor's assignment facts.
Develop a workflow	`devopsExampleAssignment.tex`, source repository materials, and authorized environment constraints.
Define evidence	`devopsExampleAssignment.tex` and controlled execution outputs.
Review a retrospective	`devopsExampleAssignment.tex` and the submitted workflow or runbook.
Edit AI guidance	`devopsAIAgentGuide.tex`, `devopsAIPromptLog.tex`, `devopsDocumentGuide.tex`, and `devopsAssignment.tex`.
Compile or troubleshoot	`devopsAssignment.tex`, `devopsDocumentGuide.tex`, the affected chapter, and the build log.

B.6 DevOps-Specific Rules

1. Prefer versioned, repeatable automation over undocumented manual steps.
2. Treat deployment output, monitoring data, and screenshots as evidence only when they reflect an actual execution.
3. Design failure exercises to be isolated, reversible, and authorized.
4. Apply least privilege and never place secrets in LaTeX, source control, prompts, screenshots, or logs.

5. State expected cloud cost, cleanup action, and resource ownership for each remote-environment activity.

6. Identify whether a recommendation is a proposal, an instructor directive, or demonstrated evidence.

B.7 Example AI Requests

B.7.1 Create an Assignment Brief

```
Read devopsAIAgentGuide.tex and use the Initialize an Assignment context.
From the supplied SSW 590 objective and constraints, fill only supported
placeholders in the introduction and activity brief. Propose clear, observable
completion criteria, a safe cleanup action, and a rubric. Do not invent cloud
access, due dates, policy, costs, or approved tools. Update history, compile,
and report instructor decisions still needed.
```

B.7.2 Review a Student Workflow

```
Read devopsAIAgentGuide.tex and the workflow, evidence, and retrospective
chapters. Review the supplied repository and sanitized evidence for
reproducibility, least privilege, observability, rollback, and cleanup.
Separate verified facts from gaps and recommendations. Do not claim that an
unexecuted command passed, and do not request credentials or operate resources.
```

B.8 Human Decisions and Information Boundaries

The instructor retains authority over assignment scope, grading, due dates, approved tooling, access, budgets, security posture, exceptions, and policy. An AI agent shall surface alternatives and evidence, not represent a decision as approved. Record material prompts, assumptions, corrections, and unresolved issues in Chapter C.

B.9 Completion Checklist

- Scope, deliverables, environment, constraints, and cleanup are clear.
- Activity identifiers and evidence references are valid and unique.
- Examples contain no secrets, personal data, or unsafe production actions.
- New controlled terms are added to the glossary and indexed.
- The newest approved change is recorded in document history.
- The complete build succeeds with no unresolved references or missing files.

- A human reviews all decisions requiring instructional or operational authority.

Appendix C

AI Prompt Log

C.1 How to Use This Appendix

Record AI assistance that materially affects wording, automation, evidence interpretation, troubleshooting, or feedback: enough context for a reviewer to see the input, output, assumptions, and human corrections. Never record credentials or personal data.

C.2 Record: IT Manual Skeleton and Gitea Chapter Outline

Date: October 12, 2026; Tool and model: Claude Code (Claude Sonnet 5.5); Task: set up this separate IT manual and prepare the Gitea chapter before any infrastructure exists. Inputs: the course manual [1] (Version 1.18, Chapters 1 and 2 read in full; it is logged here as the governing source), and the conventions of the earlier `report/` document. Output used: the manual skeleton, the Gitea chapter outline with eight rubric-aligned subsections and an extra-credit subsection, the Hosts and Passwords tables, a draft cloud-agnostic Compose configuration under `gitea/`, and a work split between the two members. The assistant created no cloud resources, executed no deployment, and recorded no test results; every evidence item is **PENDING**. The first draft of the skeleton was derived from `report/` because the course template was not yet available; it was rebuilt from the Canvas template (version 1.3) after Brahmi Dalvi supplied it, which also required merging the per-section files into the single `devopsGitea.tex` chapter file the template's AI guide requires. Human decisions recorded: SQLite is the final database choice; Brahmi Dalvi owns the cloud deployment, DNS, HTTPS, and firewall work; the 20-point backup/restore extra credit is a committed deliverable. Human review and corrections: **[TODO: BD/SD to record review of these drafts]**.

Glossary

automation A repeatable, version-controlled procedure executed by software rather than undocumented manual steps. 17

container An isolated, runnable instance of a container image that packages an application with its dependencies. 17

container image An immutable, layered template built from a Dockerfile from which containers are started. 17

deployment The controlled delivery of a versioned change to an environment. 17

Docker Compose A tool that runs a multi-container application from a versioned YAML file. 17

evidence A retained record of an actual execution, observation, or review. 17

Gitea A self-hosted Git service with issues, pull requests, and organizations. 17

Kanban board A visual workflow board with columns such as To Do, In Progress, and Complete, used to make work and work-in-progress visible. 17

observability The ability to infer system behavior from signals such as logs, metrics, traces, and health checks. 17

persistence Data surviving replacement of the container that uses it, because it lives on a mounted volume. 17

pipeline An automated sequence that builds, tests, packages, and/or deploys a change. 17

rollback A controlled return to a known acceptable version or state after a failed or harmful change. 17

runbook A documented operational procedure for a repeatable task or incident response. 17

security group A stateful virtual firewall attached to an AWS instance that lists the inbound and outbound traffic it allows. 17

synonyms Equivalent terms used in this assignment; record preferred terms to avoid ambiguity in instructions and evidence. 17

TeX Live A cross-platform distribution of TeX, LaTeX, and their packages; Overleaf compiles projects with it. 17

user data A script supplied when an EC2 instance is launched that runs once as root at first boot. 17

Acronyms

AI artificial intelligence. 17

AMI Amazon Machine Image. 17

AWS Amazon Web Services. 17

CD continuous delivery or continuous deployment. 8, 17

CI continuous integration. 17

DNS Domain Name System. 17

DORA DevOps Research and Assessment. 17

EC2 Amazon Elastic Compute Cloud. 17

IaC infrastructure as code. 17

SLO service-level objective. 17

SWOT strengths, weaknesses, opportunities, and threats. 17

TLS Transport Layer Security. 17

UML Unified Modeling Language. 17

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